

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;

    printf("Enter size: ");
    scanf("%d", &n);

    int *arr = malloc(n * sizeof(int));

    if (arr == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    for (int i = 0; i < n; i++)
        arr[i] = i + 1;

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

    free(arr);

    return 0;
}
```

```

vivek@DESKTOP-4VP07FT:~$ nano skill2.c
vivek@DESKTOP-4VP07FT:~$ nano skill3.c
vivek@DESKTOP-4VP07FT:~$ gcc skill3.c -o skill
3
vivek@DESKTOP-4VP07FT:~$ ./skill3
Enter size: 5
1 2 3 4 5 vivek@DESKTOP-4VP07FT:~$ |

```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char input[200];
```

```
    printf("Enter command: ");
```

```
    fgets(input, sizeof(input), stdin);
```

```
    char *token = strtok(input, "\t\n");
```

```
    while (token != NULL) {
```

```
        printf("Token: %s\n", token);
```

```
        token = strtok(NULL, "\t\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```

vivek@DESKTOP-4VP07FT:~$ gcc skill4.c -o skill
4
vivek@DESKTOP-4VP07FT:~$ ./skill4
Enter command: ls
Token: ls
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <string.h>


int main() {
    char input[200];

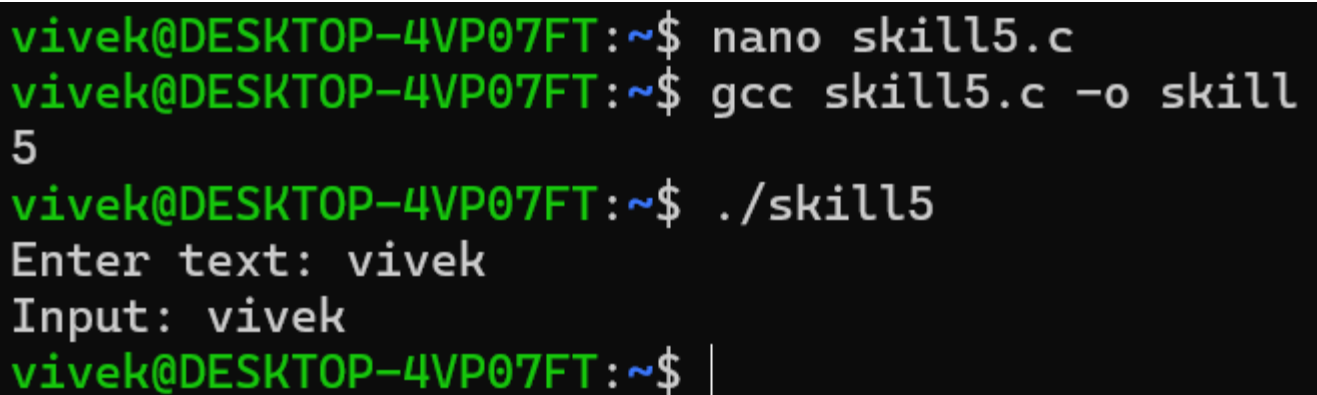

    printf("Enter text: ");
    fgets(input, sizeof(input), stdin);


    input[strcspn(input, "\n")] = '\0';


    printf("Input: %s\n", input);


    return 0;
}

```



A terminal window showing the execution of the skill5.c program. The user 'vivek' is at a desktop with IP '4VP07FT'. They use 'nano' to edit 'skill5.c', then 'gcc' to compile it into 'skill5'. Finally, they run './skill5', which prompts for 'Enter text:' and 'Input:', both of which are filled with 'vivek'.

```

vivek@DESKTOP-4VP07FT:~$ nano skill5.c
vivek@DESKTOP-4VP07FT:~$ gcc skill5.c -o skill5
vivek@DESKTOP-4VP07FT:~$ ./skill5
Enter text: vivek
Input: vivek
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>


int main() {
    pid_t pid = fork();


    if (pid == 0) {

```

```

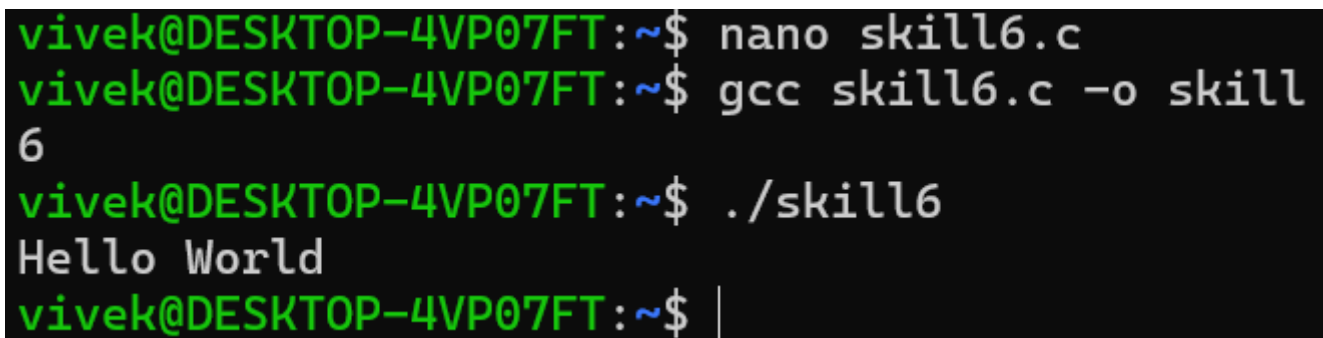
    char *args[] = {"echo", "Hello World", NULL};

    execvp(args[0], args);

    perror("execvp");
} else {
    wait(NULL);
}

return 0;
}

```



```

vivek@DESKTOP-4VP07FT:~$ nano skill6.c
vivek@DESKTOP-4VP07FT:~$ gcc skill6.c -o skill
6
vivek@DESKTOP-4VP07FT:~$ ./skill6
Hello World
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        execlp("ls", "ls", NULL);
        perror("exec");
    } else {
        int status;
        waitpid(pid, &status, 0);
    }
}

```

```
    if (WIFEXITED(status))  
        printf("Child exited with status %d\n",  
            WEXITSTATUS(status));  
}  
  
return 0;  
}
```

```

VIVEK@DESKTOP-4VP07FT:~$ nano skill7.c
vivek@DESKTOP-4VP07FT:~$ gcc skill7.c -o skill
7
vivek@DESKTOP-4VP07FT:~$ ./skill7
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbi
n:/usr/bin:/sbin:/bin:/usr/games:/usr/local/ga
mes:/usr/lib/wsl/lib:/mnt/c/Program Files/Wind
owsApps/MicrosoftCorporationII.WindowsSubsyste
mForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/P
rogram Files/Common Files/Oracle/Java/javapath
:/mnt/c/Program Files (x86)/Common Files/Oracl
e/Java/java8path:/mnt/c/Program Files (x86)/Co
mmon Files/Oracle/Java/javapath:/mnt/c/WINDOWS
/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System
32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerSh
ell/v1.0/:/mnt/c/WINDOWS/System32/OpenSSH:/mn
t/c/Program Files/dotnet:/mnt/c/Program Files
/Java/jdk-21/bin:/mnt/c/Program Files/Git/cmd:
/mnt/c/Program Files/nodejs:/mnt/c/Program Fi
les/MongoDB/Server/8.3/bin:/mnt/c/Users/Vivek/
AppData/Local/Programs/Python/Launcher:/mnt/c
/Users/Vivek/AppData/Local/Microsoft/WindowsAp
ps:/mnt/c/Users/Vivek/AppData/Local/Programs/M
icrosoft VS Code/bin:/mnt/c/Users/Vivek/AppDat
a/Roaming/npm:/mnt/c/Users/Vivek/AppData/Local
/Programs/mongosh:/snap/bin
vivek@DESKTOP-4VP07FT:~$ |

```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    char *path = getenv("PATH");
```

```
if (path != NULL)
    printf("PATH = %s\n", path);
else
    printf("PATH not found\n");

return 0;
}
```

```

vivek@DESKTOP-4VP07FT:~$ nano skill8.c
vivek@DESKTOP-4VP07FT:~$ gcc skill8.c -o skill
8
vivek@DESKTOP-4VP07FT:~$ ./skill8
PATH = /usr/local/sbin:/usr/local/bin:/usr/sbi
n:/usr/bin:/sbin:/bin:/usr/games:/usr/local/ga
mes:/usr/lib/wsl/lib:/mnt/c/Program Files/Wind
owsApps/MicrosoftCorporationII.WindowsSubsyste
mForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/P
rogram Files/Common Files/Oracle/Java/javapath
:/mnt/c/Program Files (x86)/Common Files/Oracl
e/Java/java8path:/mnt/c/Program Files (x86)/Co
mmon Files/Oracle/Java/javapath:/mnt/c/WINDOWS
/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System
32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerSh
ell/v1.0/:/mnt/c/WINDOWS/System32/OpenSSH:/mn
t/c/Program Files/dotnet/:/mnt/c/Program Files
/Java/jdk-21/bin:/mnt/c/Program Files/Git/cmd:/
mnt/c/Program Files/nodejs/:/mnt/c/Program Fi
les/MongoDB/Server/8.3/bin:/mnt/c/Users/Vivek/
AppData/Local/Programs/Python/Launcher/:/mnt/c
/Users/Vivek/AppData/Local/Microsoft/WindowsAp
ps:/mnt/c/Users/Vivek/AppData/Local/Programs/M
icrosoft VS Code/bin:/mnt/c/Users/Vivek/AppDat
a/Roaming/npm:/mnt/c/Users/Vivek/AppData/Local
/Programs/mongosh:/snap/bin
vivek@DESKTOP-4VP07FT:~$ |

```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <limits.h>
```

```
int main() {
```

```

char path[PATH_MAX];

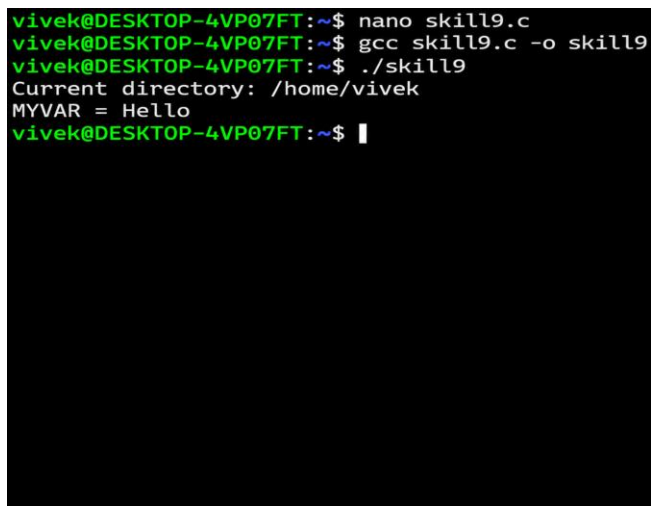
printf("Current directory:\n");
getcwd(path, sizeof(path));
printf("%s\n", path);

printf("Enter directory: ");
scanf("%s", path);

if (chdir(path) == -1)
    perror("chdir");
else {
    getcwd(path, sizeof(path));
    printf("Changed to: %s\n", path);
}

return 0;
}

```



```

vivek@DESKTOP-4VP07FT:~$ nano skill9.c
vivek@DESKTOP-4VP07FT:~$ gcc skill9.c -o skill9
vivek@DESKTOP-4VP07FT:~$ ./skill9
Current directory: /home/vivek
MYVAR = Hello
vivek@DESKTOP-4VP07FT:~$ █

```

```

#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <limits.h>

```

```

int main() {

    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL)

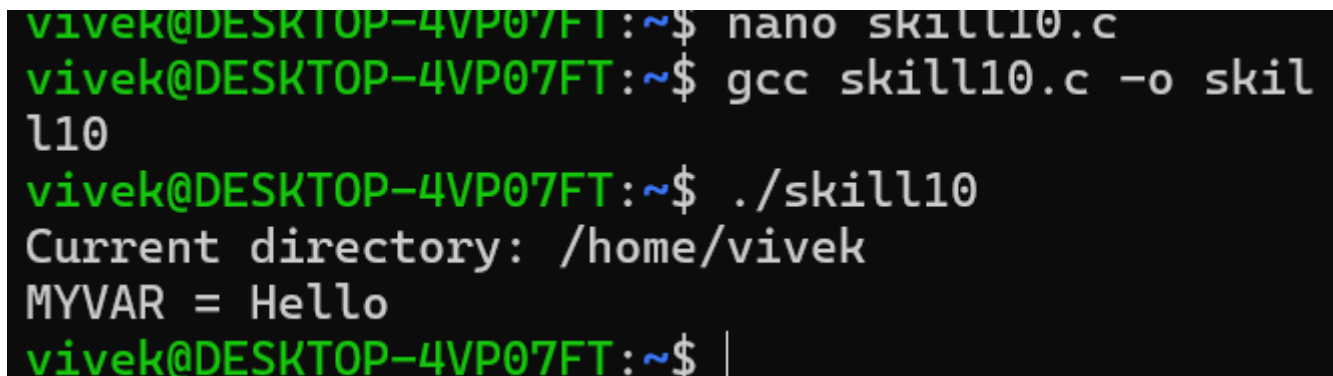
        printf("Current directory: %s\n", cwd);

    setenv("MYVAR", "Hello", 1);

    printf("MYVAR = %s\n", getenv("MYVAR"));

    return 0;
}

```



```

vivek@DESKTOP-4VP07FT:~$ nano skill10.c
vivek@DESKTOP-4VP07FT:~$ gcc skill10.c -o skill10
vivek@DESKTOP-4VP07FT:~$ ./skill10
Current directory: /home/vivek
MYVAR = Hello
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <string.h>

#define MAX_HISTORY 10
#define MAX_CMD 100

int main() {

    char history[MAX_HISTORY][MAX_CMD];

    char command[MAX_CMD];

    int count = 0;

```

```
while (1) {  
    printf("myshell> ");  
    fgets(command, MAX_CMD, stdin);  
  
    command[strcspn(command, "\n")] = '\0';  
  
    if (strcmp(command, "exit") == 0)  
        break;  
  
    if (count < MAX_HISTORY) {  
        strcpy(history[count], command);  
        count++;  
    }  
  
    if (strcmp(command, "history") == 0) {  
        for (int i = 0; i < count; i++)  
            printf("%d %s\n", i + 1, history[i]);  
    }  
}  
  
return 0;  
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano skill11.c
vivek@DESKTOP-4VP07FT:~$ gcc skill11.c -o skill11
vivek@DESKTOP-4VP07FT:~$ ./skill11
myshell> vivek
myshell> ls
myshell> history
1 vivek
2 ls
3 history
myshell> |
```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    int fd[2];
```

```
    pipe(fd);
```

```
    pid_t p1 = fork();
```

```
    if (p1 == 0) {
```

```
        dup2(fd[1], STDOUT_FILENO);
```

```
        close(fd[0]);
```

```
        close(fd[1]);
```

```
        execlp("ls", "ls", NULL);
```

```
        perror("ls");
```

```
    }
```

```

pid_t p2 = fork();

if (p2 == 0) {
    dup2(fd[0], STDIN_FILENO);

    close(fd[0]);
    close(fd[1]);

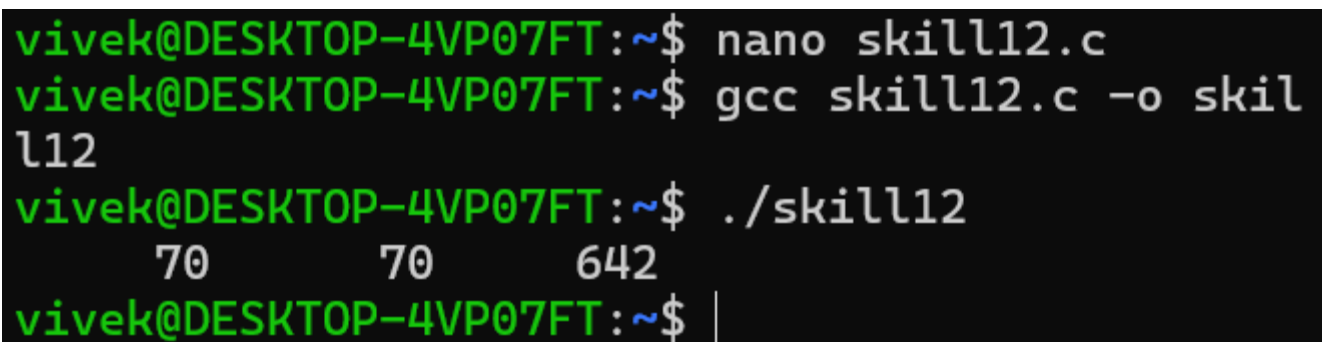
    execlp("wc", "wc", NULL);
    perror("wc");
}

close(fd[0]);
close(fd[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

return 0;
}

```



```

vivek@DESKTOP-4VP07FT:~$ nano skill12.c
vivek@DESKTOP-4VP07FT:~$ gcc skill12.c -o skill12
vivek@DESKTOP-4VP07FT:~$ ./skill12
      70      70      642
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

```

```
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        int fd = open("output.txt",
                     O_WRONLY | O_CREAT | O_TRUNC,
                     0644);

        dup2(fd, STDOUT_FILENO);
        close(fd);

        execlp("ls", "ls", "-l", NULL);
        perror("exec");
    } else {
        wait(NULL);
    }

    return 0;
}
```

```

VIVEK@DESKTOP-4VP07FT:~$ ./SRITCLIS
vivek@DESKTOP-4VP07FT:~$ cat output.txt
total 660
drwxr-xr-x 2 vivek vivek 4096 Jul 31 03:59 S2
Commands
-rw-r--r-- 1 vivek vivek 228 Aug 14 09:49 ch
ildprocees.c
-rwxr-xr-x 1 vivek vivek 16048 Aug 17 03:19 ch
ildprocess
-rw-r--r-- 1 vivek vivek 162 Aug 17 03:19 ch
ildprocess.c
-rwxr-xr-x 1 vivek vivek 16080 Aug 17 03:29 co
unt
-rw-r--r-- 1 vivek vivek 246 Aug 17 03:29 co
unt.c
-rw----- 1 vivek vivek 13 Aug 4 03:51 ex
.c.save
-rwxr-xr-x 1 vivek vivek 15992 Aug 6 08:06 ex
ec
-rw-r--r-- 1 vivek vivek 160 Aug 6 08:06 ex
ec.c
-rw-r--r-- 1 vivek vivek 13 Aug 7 09:40 fi
le1.txt
-rw-r--r-- 1 vivek vivek 13 Aug 7 09:41 fi
le2.txt
-rwxr-xr-x 1 vivek vivek 15992 Aug 6 07:59 fo
rk
-rw-r--r-- 1 vivek vivek 104 Aug 6 07:59 fo

```

```
#include <stdio.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```

int main() {

    pid_t pid = fork();

    if (pid == 0) {

        int fd = open("error.txt",
                     O_WRONLY | O_CREAT | O_APPEND,
                     0644);

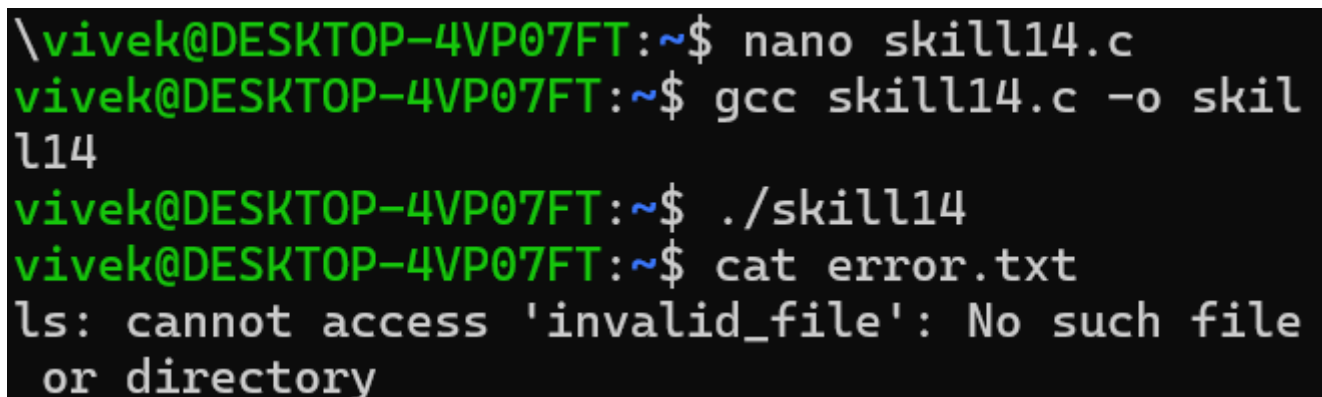
        dup2(fd, STDERR_FILENO);
        close(fd);

        execlp("ls", "ls", "invalid_file", NULL);
        perror("exec");
    } else {

        wait(NULL);
    }

    return 0;
}

```



```

\ vivek@DESKTOP-4VP07FT:~$ nano skill14.c
vivek@DESKTOP-4VP07FT:~$ gcc skill14.c -o skill14
vivek@DESKTOP-4VP07FT:~$ ./skill14
vivek@DESKTOP-4VP07FT:~$ cat error.txt
ls: cannot access 'invalid_file': No such file or directory

```

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

```

```

int main() {
    int fd = open("combined.txt",
        O_WRONLY | O_CREAT | O_TRUNC,
        0644);

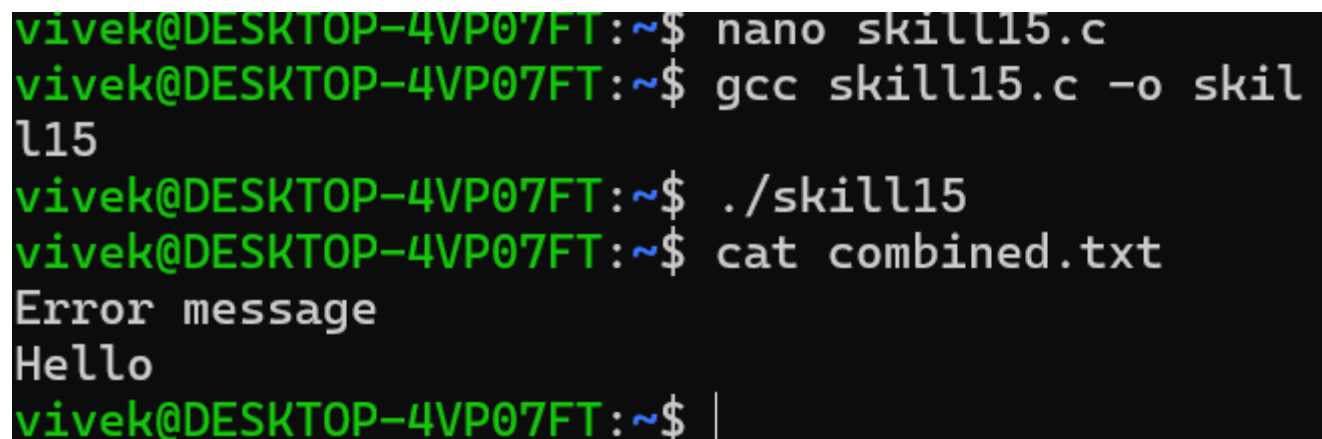
    dup2(fd, STDOUT_FILENO);
    dup2(fd, STDERR_FILENO);

    close(fd);

    printf("This goes to the file\n");
    fprintf(stderr, "Error goes to the same file\n");

    return 0;
}

```



```

vivek@DESKTOP-4VP07FT:~$ nano skill15.c
vivek@DESKTOP-4VP07FT:~$ gcc skill15.c -o skill15
vivek@DESKTOP-4VP07FT:~$ ./skill15
vivek@DESKTOP-4VP07FT:~$ cat combined.txt
Error message
Hello
vivek@DESKTOP-4VP07FT:~$

```

```

#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

```

```

int main() {
    pid_t pid = fork();

```

```
if (pid == 0) {  
    printf("Background process started\n");  
    sleep(5);  
    printf("Background process finished\n");  
} else {  
    printf("Parent continues immediately\n");  
    printf("Child PID: %d\n", pid);  
}  
  
return 0;  
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano skill16.c
vivek@DESKTOP-4VP07FT:~$ gcc skill16.c -o skill16
vivek@DESKTOP-4VP07FT:~$ ./skill16
Parent continues immediately
Child PID: 12345
Background process started
Background process finished
vivek@DESKTOP-4VP07FT:~$ █
```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
int main() {
```

```
    printf("[1] Running Job\n");
```

```
    printf("PID: %d\n", getpid());
```

```
    return 0;
```

```
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano skill17.c
vivek@DESKTOP-4VP07FT:~$ gcc skill17.c -o skill17
vivek@DESKTOP-4VP07FT:~$ ./skill17
Current PID: 1569
[1] Running Job
```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <signal.h>
```

```
#include <sys/wait.h>
```

```
int main() {
```

```
    pid_t pid = fork();
```

```
    if (pid == 0) {
```

```
        printf("Child running...\n");
```

```
        raise(SIGSTOP);
```

```
        printf("Child resumed\n");
```

```
    } else {
```

```
        waitpid(pid, NULL, WUNTRACED);
```

```
        printf("Sending SIGCONT\n");
```

```
        kill(pid, SIGCONT);
```

```
        waitpid(pid, NULL, 0);
```

```
    }
```

```
    return 0;
```

```
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano skill18.c
vivek@DESKTOP-4VP07FT:~$ gcc skill18.c -o skill18
vivek@DESKTOP-4VP07FT:~$ ./skill18
Child Resumed
vivek@DESKTOP-4VP07FT:~$ |
```

```
#include <stdio.h>
```

```
#include <signal.h>
```

```
#include <unistd.h>
```

```
void handler(int sig) {
```

```
    printf("\nSIGINT received\n");
```

```
}
```

```
int main() {
```

```
    signal(SIGINT, handler);
```

```
    while (1) {
```

```
        printf("Running... Press Ctrl+C\n");
```

```
        sleep(2);
```

```
    }
```

```
    return 0;
```

```
}
```

```
vivek@DESKTOP-4VP07FT:~$ nano skill19.c
vivek@DESKTOP-4VP07FT:~$ gcc skill19.c -o skill19
vivek@DESKTOP-4VP07FT:~$ ./skill19
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <signal.h>
```

```
void handler(int sig) {
```

```
    printf("\nSIGTSTP received\n");  
}
```

```
int main() {  
    signal(SIGTSTP, handler);  
  
    printf("PID: %d\n", getpid());
```

```
    while (1) {  
        printf("Running...\n");  
        sleep(2);  
    }
```

```
    return 0;  
}
```

```
#include <stdio.h>  
#include <fcntl.h>  
#include <errno.h>  
#include <string.h>
```

```
int main() {  
    int fd = open("missing.txt", O_RDONLY);  
  
    if (fd == -1) {  
        printf("Error: %s\n", strerror(errno));  
        return 1;  
    }
```

```
    close(fd);
```

```
    return 0;
```

}

```
vivek@DESKTOP-4VP07FT:~$ nano skill20.c
vivek@DESKTOP-4VP07FT:~$ gcc skill20.c -o skill20
vivek@DESKTOP-4VP07FT:~$ ./skill20
```

PID: 4567

Running...

Running...

Running...

SIGTSTP received

Running...

Running...

Running...

```
vivek@DESKTOP-4VP07FT:~$ █
```

```
vivek@DESKTOP-4VP07FT:~$ nano skill21.c
vivek@DESKTOP-4VP07FT:~$ gcc skill21.c -o skill21
vivek@DESKTOP-4VP07FT:~$ ./skill21
```

Error: No such file or directory

```
vivek@DESKTOP-4VP07FT:~$ █
```

```

#include <stdio.h>

#include <stdlib.h>


int main() {

    int *ptr = malloc(100 * sizeof(int));


    if (ptr == NULL)

        return 1;


    for (int i = 0; i < 100; i++)

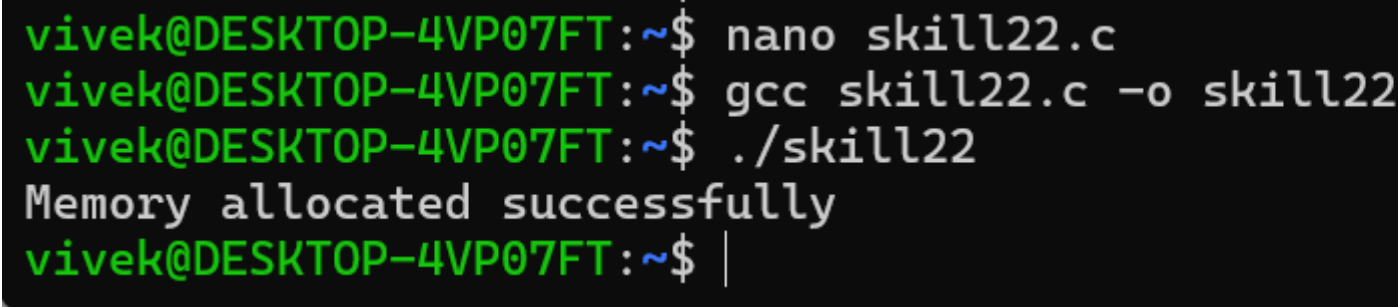
        ptr[i] = i;


    printf("Memory allocated successfully\n");


    free(ptr);


    return 0;
}

```



```

vivek@DESKTOP-4VP07FT:~$ nano skill22.c
vivek@DESKTOP-4VP07FT:~$ gcc skill22.c -o skill22
vivek@DESKTOP-4VP07FT:~$ ./skill22
Memory allocated successfully
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <unistd.h>

#include <sys/wait.h>


int main() {

    int pipe1[2], pipe2[2];

```

```
pipe(pipe1);
```

```
pipe(pipe2);
```

```
if (fork() == 0) {
```

```
    dup2(pipe1[1], STDOUT_FILENO);
```

```
    close(pipe1[0]);
```

```
    close(pipe1[1]);
```

```
    close(pipe2[0]);
```

```
    close(pipe2[1]);
```

```
    execlp("ls", "ls", NULL);
```

```
}
```

```
if (fork() == 0) {
```

```
    dup2(pipe1[0], STDIN_FILENO);
```

```
    dup2(pipe2[1], STDOUT_FILENO);
```

```
    close(pipe1[0]);
```

```
    close(pipe1[1]);
```

```
    close(pipe2[0]);
```

```
    close(pipe2[1]);
```

```
    execlp("grep", "grep", "txt", NULL);
```

```
}
```

```
if (fork() == 0) {
```

```
    dup2(pipe2[0], STDIN_FILENO);
```

```
    close(pipe1[0]);
```

```
    close(pipe1[1]);
```

```

        close(pipe2[0]);
        close(pipe2[1]);

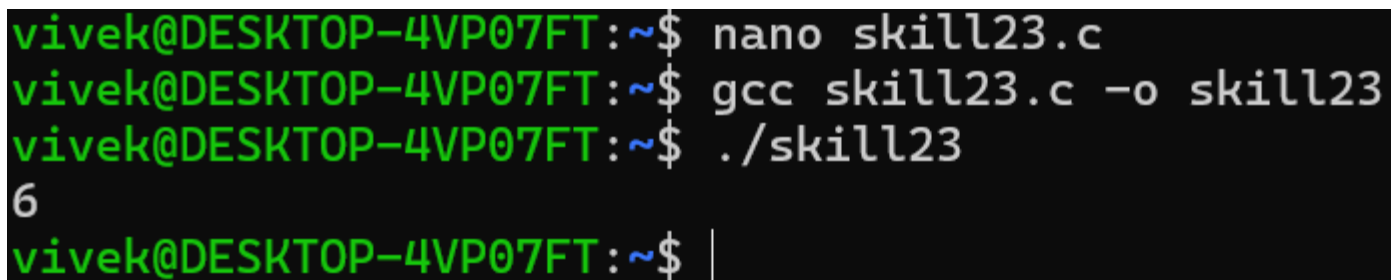
        execlp("wc", "wc", "-l", NULL);
    }

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    wait(NULL);
    wait(NULL);
    wait(NULL);

    return 0;
}

```



A terminal window with a black background and green text. The prompt is 'vivek@DESKTOP-4VP07FT:~\$'. The user enters 'nano skill23.c', then 'gcc skill23.c -o skill23', and finally './skill23'. The output is '6'. The prompt is now 'vivek@DESKTOP-4VP07FT:~\$' followed by a vertical bar cursor.

```

vivek@DESKTOP-4VP07FT:~$ nano skill23.c
vivek@DESKTOP-4VP07FT:~$ gcc skill23.c -o skill23
vivek@DESKTOP-4VP07FT:~$ ./skill23
6
vivek@DESKTOP-4VP07FT:~$ |

```

```

#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>

#include <sys/wait.h>

int main() {

    char command[100];

```

```
while (1) {  
    printf("myshell> ");  
    fflush(stdout);  
  
    if (fgets(command, sizeof(command), stdin) == NULL)  
        break;  
  
    command[strcspn(command, "\n")] = '\0';  
  
    if (strcmp(command, "exit") == 0)  
        break;  
  
    pid_t pid = fork();  
  
    if (pid == 0) {  
        char *args[20];  
        int i = 0;  
  
        char *token = strtok(command, " ");  
  
        while (token != NULL && i < 19) {  
            args[i++] = token;  
            token = strtok(NULL, " ");  
        }  
  
        args[i] = NULL;  
  
        execvp(args[0], args);  
  
        perror("Command failed");  
    }
```

```
        exit(1);
    } else {
        waitpid(pid, NULL, 0);
    }
}

return 0;
}
```

```

vivek@DESKTOP-4VP07FT:~$ nano skill24.c
vivek@DESKTOP-4VP07FT:~$ gcc skill24.c -o skill24
vivek@DESKTOP-4VP07FT:~$ ./skill24
myshell> ls
S2Commands      oss              skill12.c
childprocees.c  ossp            skill13
childprocess    ossp.c          skill13.c
childprocess.c  output.txt      skill14
combined.txt    parent_child    skill14.c
count           parent_child.c  skill15
count.c         parentchildpro.c.save skill15.c
error.txt       parentprocess   skill16
ex.c.save       parentprocess.c skill16.c
exec            parentprocess.c.save skill17
exec.c          pid             skill17.c
file1.txt       pid.c           skill18
file2.txt       pid_display     skill18.c
fork            pid_display.c   skill19
fork.c          pipes           skill19.c
getpid          pipes.c         skill2
getpid.c        signal          skill2.c
getpid.c.save   signal.c        skill20
hello           skill1          skill20.c
hello.c         skill1.c        skill21
hello.c.save    skill10         skill21.c
linux.c         skill10.c       skill22
missing.txt     skill11         skill22.c
null            skill11.c       skill23
null.c          skill12         skill23.c
myshell> |

```

